

Straight-Line Graphs

Test · 26 marks total

Name: _____

Date: _____

Score: _____ / 26

Section A — Reading gradient and y-intercept

A1. State the gradient of $y = 6x - 4$.

[1 mark]

A2. State the y-intercept of $y = 6x - 4$.

[1 mark]

A3. State the gradient of $y = -2x + 9$.

[1 mark]

A4. State the y-intercept of $y = -2x + 9$.

[1 mark]

Section B — Finding the gradient from two points

B1. Find the gradient of the line through (1, 2) and (3, 10).

[2 marks]

B2. Find the gradient of the line through (0, 7) and (4, -1).

[2 marks]

B3. Find the gradient of the line through (-3, -4) and (1, 4).

[2 marks]

Section C — Tables and equations from data

C1. Complete a table of values for $y = 3x - 2$ when $x = -1, 0, 1, 2$, and state the coordinates of each point.

[3 marks]

C2. A line passes through the points (0, -1), (1, 1), (2, 3), (3, 5). Find its equation in the form $y = mx + c$.

[3 marks]

Section D — Parallel lines and forming equations

D1. A line is parallel to $y = 4x + 1$ and passes through (0, 6). Write down its equation.

[2 marks]

D2. Are the lines $y = 3x - 5$ and $y = 3x + 2$ parallel? Give a reason.

[2 marks]

D3. A straight line passes through (2, 5) and (6, 13). Find the equation of the line in the form $y = mx + c$.

[4 marks]

D4. Does the point (3, 7) lie on the line $y = 2x + 1$? Show your working.

[2 marks]